

Software Quality Engineering

Static Analysis and Human Review of a Python Quiz App

Final report | 27 September 2026

Project	Quiz App
Repository	jordanonlycodes/SQE-Quiz-App
Branch	static-analysis
Original version	ac648a2
Baseline analysis	91c1f2e
Analyzed final version	fc8dd51
Technology	Python, PySide6, Open Trivia Database API

Submission contents: repository with preserved Git history, Pylint evidence and this PDF report.

1. Executive Summary

The existing desktop Quiz App integrates PySide6, an OpenTDB API client, background question loading, and automated tests. Initial complete-project Pylint analysis scored 4.49/10 with 695 findings. After targeted code improvements and documented configuration, the final run scored 7.42/10 with 434 findings, a reduction of 261 (37.6%). Errors fell by 75 and warnings by 171. These figures show measurable improvement, but a remaining finding is not automatically a defect. GUI execution and the full pytest suite remain unverified in the reported environment because PySide6, pytest-mock and pytest-qt were unavailable.

2. Project Selection and Baseline Preservation

Selection evidence in deliverable/deliverable1.md describes 1,454 source LOC, 2,495 test LOC, and 39 Python source/test files. The application has multiple modules and classes, external API integration, GUI and worker components, and both pytest and unittest suites. This scale and variety make it suitable for static analysis and manual review. The original source is recoverable at commit ac648a2. Commit 91c1f2e records the baseline analysis before improvement commits. The original Git objects and history are included in the submission repository.

3. Initial Pylint Baseline

Category	Baseline
Score	4.49/10
Total	695
Errors	90
Warnings	323
Refactoring	26
Convention	256

Significant locations include main.py, modules/gui/main_window.py, modules/questions/models.py, modules/questions/opentdb_client.py, GUI worker/overlay components, and backend test modules. The complete initial console evidence is deliverable/pylint_baseline.txt. The baseline JSON is preserved at repository root as pylint_baseline.json.

4. Quality-Oriented Classification

Problem type	Representative messages	Review interpretation
Object lifecycle and design	W0201, R0902, R0913, R0917	Inspect initialization and cohesion; thresholds alone are not proof of a bug.
Error handling and control flow	W0718, R1720	Consider GUI boundary behavior and simplify validation flow.
Portability and API usage	W1514, C0123, W0622	Actionable encoding, type-check and naming issues.
Tests and framework interaction	R0801, W0212, W0621, E0401/E0611	Determine deliberate test patterns and missing dependencies.
Formatting and documentation	C0301/C0303/C0304, C0114/C0115/C0116	Prioritize based on impact; avoid confusing volume with severity.

Decision scheme: A = fix now; B = context dependent and review; C = acceptable with explicit justification. The ten investigated items below contain five A and five B decisions. No one of these ten is classified C; C remains available for individually justified low-risk findings elsewhere. The final

configuration disables only the named protected-access check, not warning/error/refactoring categories.

5. Investigation of Ten Baseline Findings

Line numbers and examples below refer to the original commit ac648a2 and are cross-checked against deliverable/pylint_baseline.txt. Later edits shift line numbers. Each item identifies the quality impact and the A/B/C decision.

1. W0201 - attribute-defined-outside-init | A

Location: modules/gui/main_window.py:221. Pylint message: Attribute '_close_requested' defined outside __init__ (attribute-defined-outside-init)

```
26     def __init__(self) -> None:
27         super().__init__()
28         self._setup_window()
29         self._setup_layout()
30         self._init_default_variables()
31         self._setup_menu_bar()
32         self._setup_error_overlay()
33         self._setup_loading_overlay()
34         self._setup_initial_widget()

48     def _init_default_variables(self) -> None:
49         # List of ignored widgets, from MainWindow clean method
50         # Used for overlays
51         self._ignored = []
52         self._close_requested = False
```

Problem, quality impact and decision: The constructor delegated state creation to helpers; declaration outside __init__ obscured object invariants. Maintainability and testability suffer. Moved initial state into __init__ (2c5022d).

2. W0718 - broad-exception-caught | B

Location: modules/gui/main_window.py:112. Pylint message: Catching too general exception Exception (broad-exception-caught)

```
107         self._thread_controller = WorkerThreadController(self._question_loader)
108
109         self._connect_question_loading_signals()
110         self._thread_controller.setup()
111         self._thread_controller.run_thread()
112     except Exception:
113         logger.exception("Failed to start question loading.")
114         self._on_error(RuntimeError("Unexpected application error. Please try
again."))
```

Problem, quality impact and decision: A broad catch can hide programmer errors. At a GUI/worker boundary it can also keep the interface responsive and log failures. Retain with logging and error signaling; monitor whether exceptions are swallowed.

3. R1720 - no-else-raise | A

Location: modules/questions/opentdb_client.py:82. Pylint message: Unnecessary "elif" after "raise", remove the leading "el" from "elif" (no-else-raise)

```
79         logger.debug("OpenTDB response status: %s", response.status_code)
80         response.raise_for_status()
81         data = response.json()
82         if not data['results']:
83             raise NoQuestionsFoundError('No questions found for selected
parameters. Try different category, difficulty or type.')
84         elif int(amount) > len(data['results']):
85             raise NotEnoughQuestionsError('Not enough questions found for the
selected parameters. Try lower number.')
```

Problem, quality impact and decision: The first condition raises; an elif adds needless nesting and suggests dependence. Changed to a separate if (2c5022d), improving clarity.

4. R0913 - too-many-arguments | B

Location: modules/questions/models.py:13. Pylint message: Too many arguments (8/5)
(too-many-arguments)

```

10 class Question:
11     """Data object for one quiz question and its scoring data."""
12
13     def __init__(
14         self, tp: str,
15         difficulty: str,
16         category: str,
17         question: str,
18         correct_answer: str,
19         incorrect_answers: list[str],
20         all_answers: list[str],
21         points: int,
22     ) -> None:

```

Problem, quality impact and decision: Question takes eight data fields. This increases call-site burden, but each field describes one question. Consider a named constructor only if callers become error prone.

5. R0917 - too-many-positional-arguments | B

Location: modules/questions/models.py:13. Pylint message: Too many positional arguments (8/5)
(too-many-positional-arguments)

```

13     def __init__(
14         self, tp: str,
15         difficulty: str,
16         category: str,
17         question: str,
18         correct_answer: str,
19         incorrect_answers: list[str],
20         all_answers: list[str],
21         points: int,
22     ) -> None:
23         self.tp = tp
24         self.difficulty = difficulty
25         self.category = category

```

Problem, quality impact and decision: Eight positional fields are easy to transpose. The data model explains the shape; keyword calls or a factory are possible future improvements if usage expands.

6. R0902 - too-many-instance-attributes | B

Location: modules/gui/main_window.py:14. Pylint message: Too many instance attributes (11/7)
(too-many-instance-attributes)

```

14 class MainWindow(QMainWindow):
15     """
16     Main application window and top-level GUI coordinator.
17
18     MainWindow owns the central stacked layout, menu bar, loading/error overlays,
19     and the question-loading worker controller. It coordinates navigation between
20     the welcome screen, start screen, and question display.
21
22     Question loading is delegated to a worker thread. The OpenTDB request timeout
23     is configured in OpenTriviaClient.
24     """
25
26     def __init__(self) -> None:
27         super().__init__()

```

```

28         self._setup_window()
29         self._setup_layout()
30         self._init_default_variables()
31         self._setup_menu_bar()
32         self._setup_error_overlay()
33         self._setup_loading_overlay()
34         self._setup_initial_widget()

```

Problem, quality impact and decision: MainWindow coordinates overlays, navigation, menus and worker state. Eleven attributes signal a review opportunity; they do not alone prove poor cohesion. Decompose when a stable separate responsibility emerges.

7. R0801 - duplicate-code | B

Location: tests/unittest_backend/test_opentdb_client_unittest.py:1. Pylint message: Similar lines in 2 files

```

84         @patch("modules.questions.opentdb_client.requests.get")
85         def test_get_questions_data_raises_error_when_no_questions(self,
mock_requests_get: MagicMock) -> None:
86             """Test that an empty results list raises OpenTriviaClientError."""
87             mock_response = Mock()
88             mock_response.status_code = 200
89             mock_response.json.return_value = {
90                 "response_code": 0,
91                 "results": []
92             }
93             mock_requests_get.return_value = mock_response

```

Problem, quality impact and decision: Similar backend cases appear in unittest and pytest. The duplication costs maintenance but may be intentional coverage of two test approaches; retain pending a test-suite policy decision.

8. W1514 - unspecified-encoding | A

Location: main.py:102. Pylint message: Using open without explicitly specifying an encoding (unspecified-encoding)

```

100         # Load and apply styles from styles.css to the entire application
101         stylesheet_path = "modules/gui/styles/styles.css"
102         with open(stylesheet_path, "r") as file:
103             app.setStyleSheet(file.read())
104         logger.debug("App stylesheet loaded from: %s", stylesheet_path)

```

Problem, quality impact and decision: Platform-default stylesheet decoding can differ between machines. Added encoding="utf-8" to the open call (2c5022d).

9. C0123 - unidiomatic-typecheck | A

Location: modules/gui/widgets/overlays/error_overlay.py:62. Pylint message: Use isinstance() rather than type() for a typecheck. (unidiomatic-typecheck)

```

60         """
61         label_text = str(error)
62         if type(error) is OpenTriviaClientError:
63             label_text += " Try again in a few seconds."
64         self._error_label.setText(label_text)
65         self._error_label.update()

```

Problem, quality impact and decision: Exact type comparison excludes subclasses and is less idiomatic. Replaced with isinstance(error, OpenTriviaClientError) (2c5022d).

10. W0622 - redefined-builtin | A

Location: modules/gui/menu_bar/menus/quiz_menu.py:38. Pylint message: Redefining built-in 'exit' (redefined-builtin)

```

37         def _add_exit_action(self) -> None:

```

```

38         exit = QAction('Exit', self)
39         exit.setShortcut('Ctrl+Q')
40         exit.triggered.connect(self._on_exit_action_triggered)
41         self.addAction(exit)
42

```

Problem, quality impact and decision: The local name `exit` shadows a built-in and obscures its role. Renamed it `exit_action` (2c5022d).

6. Manual Review (Separate from Pylint Findings)

Question data transformation

`modules/questions/models.py`. Independent human finding: extracting API fields, decoding HTML, constructing answer lists, randomizing, scoring and creating objects were combined. Split into smaller helpers in f25825a; this improves cohesion.

Repeated parameter-frame construction

`modules/gui/widgets/start_display/components/question_params_widget.py`. Independent human finding: four similar frame-construction sequences were consolidated into a helper in f25825a. This reduces synchronized edits.

MainWindow responsibility span

`modules/gui/main_window.py`. Human interpretation of R0902: navigation, overlays, worker lifecycle and shutdown share one coordinator. Future decomposition should follow actual responsibilities.

Startup responsibilities

`main.py`. Independent human finding: working directory, logging, `QApplication`, font, stylesheet and window setup are combined. A later bootstrap separation would improve focused tests.

Dual test frameworks

`tests/pytest_backend` and `tests/unittest_backend`. Human interpretation of R0801: overlapping cases in `pytest` and `unittest` impose maintenance cost but can intentionally demonstrate both frameworks.

7. Pylint Configuration and Rationale

Setting in <code>.pylintrc</code>	Rationale
<code>extension-pkg-allow-list=PySide6</code>	Allows inspection of extension-backed PySide6 APIs and reduces environment-related false positives.
<code>disable=protected-access</code>	Tests intentionally inspect internal GUI/model state; this one named check is suppressed project-wide.
<code>good-names=tp</code>	Accepts a short domain name for type in question/API data.
<code>max-line-length=100</code>	Keeps a 100-character formatting standard; it does not relax the limit.

The baseline and final runs use different code/configuration states, so score change cannot be attributed solely to fixes. In particular, suppressed `protected-access` diagnostics affect counts. Broad error, warning and refactoring groups were not disabled.

8. Implemented Code Improvements

At least five significant issues were addressed: W0201, R1720, W1514, C0123 and W0622. Structural changes include explicit `MainWindow` state initialization, decomposition of question transformation into focused helpers, and extraction of repeated parameter-frame setup. Other changes included

UTF-8 stylesheet decoding. Commits f25825a and 2c5022d preserve separate improvement steps.

9. Before and After Analysis

Metric	Baseline	Final	Change
Pylint score	4.49/10	7.42/10	+2.93
Total findings	695	434	-261
Errors	90	15	-75
Warnings	323	152	-171
Refactoring	26	13	-13
Convention	256	254	-2
Fatal	-	0	-

Total findings decreased by approximately 37.6%. Errors, warnings and refactoring findings decreased substantially while convention findings changed only slightly. Counts reflect both code edits and configuration; neither a score nor a raw message count measures correctness directly. Final evidence: deliverable/pylint_final.txt and deliverable/pylint_final.json (434 entries).

10. Verification and Scope

Check	Recorded result
<code>python -m compileall -q main.py modules tests</code>	Passed
<code>python -m unittest discover -s tests/unittest_backend -p 'test_*.py'</code>	19 tests run; 19 passed
Final project-wide Pylint	Completed; score 7.42/10, 434 findings, fatal 0
Full GUI pytest suite	Could not complete: PySide6, pytest-mock and pytest-qt unavailable
Application launch	Blocked: PySide6 unavailable

Passing compilation and backend tests supports the edited code paths only within their coverage. No GUI test pass or successful application launch is claimed.

11. Static Analysis and Human Review

Pylint consistently flags suspicious patterns at scale, such as undefined imports in an environment without dependencies, constructor size thresholds and duplicate blocks. Human review checks whether a warning is actionable in context, how the GUI handles errors, and whether a long transformation has mixed responsibilities. A false positive or context-dependent finding may arise from PySide6 dynamic APIs, deliberate test access or duplicated test styles. Static analysis cannot establish runtime GUI behavior, API reliability, or whether the user experience is correct. It guides review rather than replacing it.

12. Conclusion and Reproducibility

The project now has an auditable baseline, ten investigated findings, five manual review observations, targeted fixes, a documented Pylint configuration and final comparison. The measured improvement is substantial while remaining findings require individual review. The preserved history is: ac648a2 original; 91c1f2e baseline; f25825a transformation refactor; 2c5022d quality fixes; ea96954 Pylint configuration and baseline evidence; 5449e69 selection evidence; 4ea4005 initial report; fc8dd51 final analysis evidence. A later report-only commit may follow in the packaged repository.

```
git checkout static-analysis
python -m compileall -q main.py modules tests
python -m unittest discover -s tests/unittest_backend -p "test_*.py"
# Windows: deliverable\run_final_analysis.ps1
```

Appendix A. Baseline Evidence Index

Authoritative raw baseline files: deliverable/pylint_baseline.txt and repository-root/pylint_baseline.json; analyzed source: ac648a2; score 4.49/10; 695 findings. The selected verbatim messages below are excerpts; consult the raw files for every finding and detailed file distribution.

```
modules\gui\main_window.py:221:12: W0201: Attribute '_close_requested' defined outside
__init__ (attribut

modules\gui\main_window.py:112:15: W0718: Catching too general exception Exception
(broad-exception-caugh

modules\questions\opentdb_client.py:82:12: R1720: Unnecessary "elif" after "raise",
remove the leading "e

modules\questions\models.py:13:4: R0913: Too many arguments (8/5) (too-many-arguments)

modules\questions\models.py:13:4: R0917: Too many positional arguments (8/5)
(too-many-positional-argumen

modules\gui\main_window.py:14:0: R0902: Too many instance attributes (11/7)
(too-many-instance-attributes

tests\unittest_backend\test_opentdb_client_unittest.py:1:0: R0801: Similar lines in 2
files

main.py:102:9: W1514: Using open without explicitly specifying an encoding
(unspecified-encoding)

modules\gui\widgets\overlays\error_overlay.py:62:11: C0123: Use isinstance() rather than
type() for a typ

modules\gui\menu_bar\menus\quiz_menu.py:38:8: W0622: Redefining built-in 'exit'
(redefined-builtin)

Your code has been rated at 4.49/10
```

Appendix B. Final Evidence Index

Authoritative final artifacts: deliverable/pylint_final.txt and deliverable/pylint_final.json; analyzed final commit: fc8dd51. JSON contains 434 entries: 15 errors, 152 warnings, 13 refactoring and 254 convention; no fatal entries. The full text report records 7.42/10. Consult both files in the packaged repository for all individual findings.

```
Your code has been rated at 7.42/10
```